

DP024

Physics 2

Semester 2

Session

2023/2024

2 hours

SET β



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**PROGRAM DRAW A QUESTION (DAQ)
MODEL PAPER SEMESTER 2**

PHYSICS UNIT

PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI

DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

INSTRUCTIONS TO CANDIDATE:

The question paper consists of 6 questions.

Answer **all** questions.

The use of electronic calculator is permitted.

QUESTION	MARKS
1	/16
2	/12
3	/10
4	/14
5	/14
6	/14
TOTAL	/80

This question paper consists of 10 printed pages

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum <i>Laju cahaya dalam vakum</i>	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space <i>Ketelapan ruang bebas</i>	μ_o	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space <i>Ketelusan ruang bebas</i>	ϵ_o	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant <i>Pemalar Planck</i>	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass <i>Jisim neutron</i>	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass <i>Jisim proton</i>	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass <i>Jisim deuteron</i>	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant <i>Pemalar gas molar</i>	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant <i>Pemalar Avogadro</i>	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant <i>Pemalar Boltzmann</i>	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Free-fall acceleration <i>Pecutan jatuh bebas</i>	g	$= 9.81 \text{ m s}^{-2}$

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Atomic mass unit <i>Unit jisim atom</i>	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$
Electron volt <i>Elektron volt</i>	1 eV	$= 1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law <i>Pemalar hukum Coulomb</i>	$k = \frac{1}{4\pi\epsilon_0}$	$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure <i>Tekanan atmosfera</i>	1 atm	$= 1.013 \times 10^5 \text{ Pa}$
Density of water <i>Ketumpatan air</i>	ρ_w	$= 1000 \text{ kg m}^{-3}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

- | | |
|---|---|
| 1. $x = A \sin \omega t$ | 18. $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$ |
| 2. $\omega = \frac{2\pi}{T} = 2\pi f$ | 19. $C_{eff} = C_1 + C_2 + C_3 + \dots + C_n$ |
| 3. $v = A\omega \cos \omega t$ | 20. $\tau = RC$ |
| 4. $a = -A\omega^2 \sin \omega t$ | 21. $Q = Q_0 e^{\frac{-t}{RC}}$ |
| 5. $k = \frac{2\pi}{\lambda}$ | 22. $Q = Q_0 \left(1 - e^{\frac{-t}{RC}} \right)$ |
| 6. $v = f\lambda$ | 23. $B = \frac{\mu_o I}{2\pi r}$ |
| 7. $y(x, t) = A \sin(\omega t \pm kx)$ | 24. $B = \frac{\mu_o NI}{L}$ |
| 8. $v = \frac{dy}{dt}$ | 25. $B = \frac{\mu_o I}{2r}$ |
| 9. $y = 2A \cos kx \sin \omega t$ | 26. $\phi = BA \cos \theta$ |
| 10. $v = \sqrt{\frac{T}{\mu}}$ | 27. $\Phi = N\phi$ |
| 11. $I = \frac{dQ}{dt}$ | 28. $\varepsilon = -\frac{d\phi}{dt}$ |
| 12. $Q = ne$ | 29. $\varepsilon = Blv \sin \theta$ |
| 13. $\rho = \frac{RA}{l}$ | 30. $\varepsilon = -NA \frac{dB}{dt}$ |
| 14. $R_{eff} = R_1 + R_2 + R_3 + \dots + R_n$ | 31. $\varepsilon = -NB \frac{dA}{dt}$ |
| 15. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$ | 32. $R = 2f$ |
| 16. $V = IR$ | |
| 17. $C = \frac{Q}{V}$ | |

$$33. \quad \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$34. \quad m = \frac{h_i}{h_o} = -\frac{v}{u}$$

$$35. \quad y_m = \frac{m\lambda D}{d}$$

$$36. \quad y_m = \frac{(m + \frac{1}{2})\lambda D}{d}$$

$$37. \quad \Delta y = \frac{\lambda D}{d}$$

1. (a) The equation of a progressive wave is expressed by

$$y = 6.0 \sin (2t - x)$$

where x and y are in metres and t is in s. Calculate the

- (i) frequency and wavelength.
- (ii) particle's vibrational velocity at $t = 0$ s and $x = 0.4$ m.
- (iii) wave speed and state the direction of the wave.

[8 marks]

- (b) A standing wave is produced in a stretched string with 2.5 m length and 20 g of mass. The equation of wave is given by

$$y = 5.0 \cos(5x) \sin(2\pi t)$$

where y and x are in meters and t is in s. Calculate the

- (i) speed of the wave.
- (ii) amplitude of the standing wave at $x = 5.0$ cm.
- (iii) tensional force acting on the string.

[8 marks]

2. (a) A copper wire carries a current of 7.0 A. Calculate the amount of charge flows through a cross-sectional area of the wire in 40 s.

[1 mark]

- (b) A current of 3.0 A flows through a wire of 5.0 m long and 8 mm^2 area. The potential difference between the ends of the wire is 12 V. Calculate the

- (i) resistivity and
- (ii) new length if the resistivity is doubled.

[5 marks]

(c)

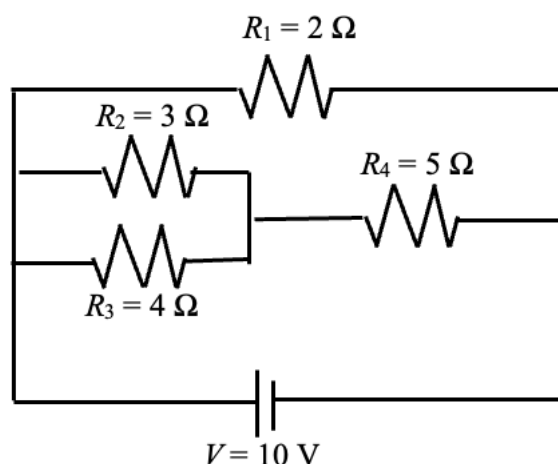
**FIGURE 2**

FIGURE 2 shows an arrangement of four resistors with $R_1 = 2\ \Omega$, $R_2 = 3\ \Omega$, $R_3 = 4\ \Omega$ and $R_4 = 5\ \Omega$ connected to a 10 V battery. Calculate the

- (i) effective resistance of the circuit.
- (ii) total current in the circuit.

[6 marks]

3. (a)

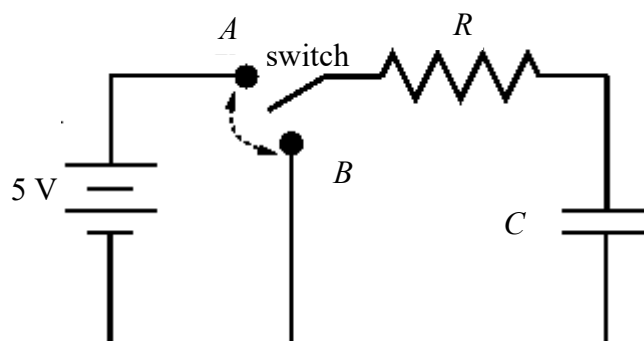
**FIGURE 3.1**

FIGURE 3.1 shows a circuit consisting of a capacitor, a resistor, a switch and a power supply. The capacitor has capacitance, $C = 500\ \mu\text{F}$, the resistor of resistance $R = 50\ \text{k}\Omega$ and the power supply, $V = 5\ \text{V}$. The capacitor starts to charge when the switch is connected to A.

- (i) Calculate the time constant.
- (ii) Calculate the time taken for the charge to increase to 25% of its maximum charge.
- (iii) Sketch a graph of charge against time for the charging process.

[6 marks]

(b)

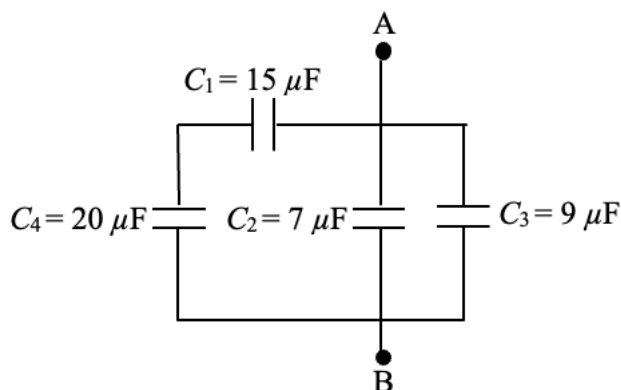
**FIGURE 3.2**

FIGURE 3.2 shows an arrangement of four capacitors $C_1 = 15 \mu\text{F}$, $C_2 = 7 \mu\text{F}$, $C_3 = 9 \mu\text{F}$ and $C_4 = 20 \mu\text{F}$ between point A and B. The potential difference across points A and B is 15 V . Calculate the

- (i) effective capacitance.
- (ii) total charge in the circuit.

[4 marks]

4. (a) A long straight wire carries a current of 50 A in the upward direction.

- (i) Sketch the magnetic field lines around the wire.
- (ii) Calculate the magnitude of the magnetic field at a point of 10 cm from the wire.

[4 marks]

(b) A solenoid with 200 turns, has a length of 0.55 m and radius of 4.8 cm. A current of 6.4 A flows through this solenoid. Calculate the

- (i) magnetic field at the centre of the solenoid.
- (ii) maximum magnetic flux produced by the solenoid.

[5 marks]

(c)

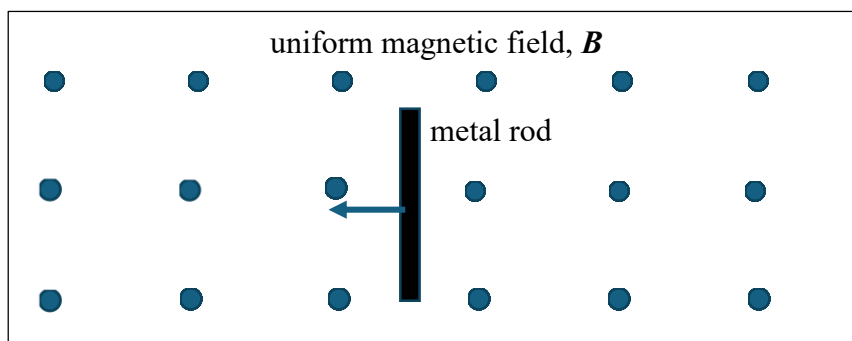
**FIGURE 4**

FIGURE 4 shows the motion of a metal rod in a uniform magnetic field, $B = 500 \text{ mT}$. The metal rod has a length of 40.0 cm and moves to the left at a constant speed of 50 m s^{-1} perpendicularly to the magnetic field. Determine the

- (i) induced emf in the metal rod.
- (ii) magnitude and direction of induced current if the rod is connected to a resistor of resistance 30Ω .

[5 marks]

5. (a) An object 0.6 cm tall is placed 16.5 cm from a concave mirror having a radius of curvature of 22.0 cm .

- (i) Sketch a ray diagram.
- (ii) Calculate the image distance.
- (iii) Calculate the height of the image.
- (iv) State three characteristics of the image formed by the mirror.

[11 marks]

- (b) An object of 4.0 mm height is placed in front of a converging lens with a focal length of 70.0 cm . If the height of upright image formed is 1.3 cm , calculate the object distance.

[4 marks]

6. (a) A two-slit interference experiment in which the slits are 0.2 mm apart and the screen is 1.0 m from the slits. The 3rd bright fringe from the central bright fringe is seen at 9.49 mm. Calculate the

- (i) wavelength of the light
- (ii) distance between the 6th dark fringe and the 7th bright fringe.

[8 marks]

(b) A screen is separated from a double-slit source by 1.20 m. The distance between the two slits is 0.03 mm and a monochromatic light of wavelength 550 nm is used.

- (i) Calculate the distance between adjacent fringes.
- (ii) Is there any changes with the fringes seen on the screen when the wavelength from the source is changed to 450 nm? Justify your answer by calculation.

[6 marks]